Categories, Proofs, and Processes:
Lecture 17 Adjunctions

Wira Azmoon Ahmad

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7 Adjunctions

7.1 Unit

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, phantom, "\perp"] & \mathcal{D} \arrow[l, bend left, "G"] \end{tikzcd}$$

is an adjunction iff $\forall X \in \mathrm{Ob}(\mathcal{C})$, there exists a unit

$$\eta_X : X \to GF(X)$$

such that

$$\forall f : X \to G(Y), \exists! \hat{f} : F(X) \to Y$$

where

$$\begin{tikzcd} X \arrow[r, "\eta_X"] \arrow[dr, "f"'] & GF(X) \arrow[d, dashed, "G(\hat{f})"] \\ & G(Y) \end{tikzcd}$$

Example 7.1. Makes more sense as unit, unit version.

$$\begin{tikzcd} \mathbf{Set} \arrow[r, bend left, "\mathbb{R}[-]"] \arrow[r, phantom, "\perp"] & \mathbf{Vect}_{\mathbb{R}} \arrow[l, bend left, "U"] \end{tikzcd}$$

$\mathbb{R}[X]$ vector space spanned by X.

$$v \in \mathbb{R}[X], v = \sum_i r_i x_i, \quad r_i \in \mathbb{R}, x_i \in X$$

$$\mathbb{R}[f] : \mathbb{R}[X] \to \mathbb{R}[Y]$$

$$\mathbb{R}[f]\left(\sum_i r_i x_i\right) = \sum_i r_i f(x_i)$$

$$
\begin{tikzcd}
b \arrow[rrr, mapsto] \arrow[dddrrr, mapsto, bend right=10] & & & b \arrow[ddd, mapsto] \\
& B \arrow[r] \arrow[dr, "f"'] & U(\mathbb{R}[B]) \arrow[d, dashed, "U(\hat{f})"] & \\
& & U(V) & \\
& & & f(b)
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbb{R}[B] \arrow[d, dashed, "\hat{f}"'] & \sum_i r_i k_i \arrow[d, mapsto] \\
V & \sum_i r_i f(b_i)
\end{tikzcd}
$$

Example 7.2. Makes more sense as co-unit, unit version.

$$(- \times B) : \mathbf{Set} \to \mathbf{Set}$$

$$\begin{cases} A \mapsto A \times B \\ f \mapsto f \times 1_B \end{cases}$$

$$g : X \to Y$$

$$B \Rightarrow g : B \Rightarrow X \to B \Rightarrow Y$$

$$(B \Rightarrow g)(h) := g \circ h$$

$$
\begin{tikzcd}
x \arrow[rrr, mapsto] \arrow[dddrr, mapsto, bend right=15] & & & \lambda b.(x, b) \arrow[ddd, mapsto] \\
& X \arrow[r, "\eta_X"] \arrow[dr, "f"'] & B \Rightarrow (X \times B) \arrow[d, dashed, "(f\circ -)"] & \\
& & B \Rightarrow Y & \\
& & & f(x) = (\hat{f} \circ -)(\lambda b.(x, b))
\end{tikzcd}
$$

$$\hat{f} : X \times B \to Y$$

$$\hat{f}(x, b) = f(x)(b)$$

$$(\hat{f} \circ -)(\lambda b.(x, b)) = \lambda b.\hat{f}(x, b) = \lambda b.f(x)(b) = f(x)$$

Last equality by η expansion.

7.1.1 Notes

- Example: X is set, $GF(X)$ is a list.
- More useful for the left adjoints.

7.2 Bijection

$$\mathcal{D}(F(X), Y) \cong \mathcal{C}(X, G(Y))$$

7.3 Co-Unit

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, phantom, "\perp"] & \mathcal{D} \arrow[l, bend left, "G"] \end{tikzcd}$$

is an adjunction iff $\forall X \in \mathrm{Ob}(\mathcal{C})$, there exists a <u>co-unit</u>

$$\varepsilon_Y : FG(Y) \to Y$$

such that

$$\forall f : F(X) \to Y, \exists! \tilde{f} : X \to G(Y)$$

where

$$\begin{tikzcd} F(X) \arrow[d, dashed, "F(\tilde{f})"'] \arrow[dr, "f"] & \\ FG(Y) \arrow[r, "\varepsilon_Y"'] & Y \end{tikzcd}$$

Note 7.3. Co-unit is some kind of evaluation.

Example 7.4. Makes more sense as unit, co-unit version.

$$F = \mathbb{R}[-], G = U$$

$$\begin{tikzcd} \mathbf{Set} \arrow[r, bend left, "\mathbb{R}[-]"] \arrow[r, phantom, "\perp"] & \mathbf{Vect}_{\mathbb{R}} \arrow[l, bend left, "U"] \end{tikzcd}$$

"Forget that Y is a vector space, treat it as a set"

$$\begin{tikzcd} \mathbb{R}[X] \arrow[d, dashed, "F(\tilde{f})"'] \arrow[dr, "f"] & \\ \mathbb{R}[U(Y)] \arrow[r, "\varepsilon_Y"'] & Y \\ \sum_i r_i \cdot [v_i] \arrow[u, rightsquigarrow] \arrow[r, mapsto] & \sum_i r_i v_i \end{tikzcd}$$

$$\tilde{f}(x) = f(x)$$

Example 7.5. Makes more sense as co-unit.

$$(- \times B) : \mathbf{Set} \to \mathbf{Set}$$

$$\begin{cases} A \mapsto A \times B \\ f \mapsto f \times 1_B \end{cases}$$

$$\mathbf{Set}(A \times B, C) \cong \mathbf{Set}(A, B \Rightarrow C)$$

$$\begin{tikzcd}
\mathbf{Set} \arrow[r, bend left, "(-\times B)"] \arrow[r, phantom, "\perp"] & \mathbf{Set} \arrow[l, bend left, "G=(B\Rightarrow -)"]
\end{tikzcd}$$

$$FG(C) = (B \Rightarrow C) \times B$$

$$\begin{tikzcd}
(a,b) \arrow[ddd, mapsto] \arrow[dddrrr, mapsto, bend left] & & & \\
& A \times B \arrow[d, dashed, "\tilde{f}\times 1_B"'] \arrow[dr, "f"] & & \\
& (B \Rightarrow C) \times B \arrow[r, "\varepsilon_C"'] & C & \\
(\lambda x. f(a,x), b) \arrow[rrr, mapsto] & & & f(a,b)
\end{tikzcd}$$

$$\tilde{f} : A \to (B \Rightarrow C)$$

$$\tilde{f} :: a \mapsto \lambda x. f(a,x)$$

$$\tilde{f} :: a \mapsto f(a,-)$$

Notation mapping of the same thing.